



Cryptography

Encryption is used on the Internet to transmit data, such as payment information, e-mails, or personal data, confidentially and protected against manipulation. Data is encrypted using various cryptographic algorithms based on mathematical operations. With the help of encryption, data can be transformed into a form that unauthorized persons can no longer read. Digital keys in `symmetric` or `asymmetric` encryption processes are used for encryption. It is easier to crack cipher texts or keys depending on the encryption methods used. If state-of-the-art cryptographic methods with extensive key lengths are used, they work very securely and are almost impossible to compromise for the time being. In principle, we can distinguish between `symmetric` and `asymmetric` encryption techniques. Asymmetric methods have only been known for a few decades. Nevertheless, they are the most frequently used methods in digital communication.

Symmetric Encryption

Symmetric encryption, also known as secret key encryption, is a method that uses the same key to encrypt and decrypt the data. This means the sender and the receiver must have the same key to decrypt the data correctly.

If the secret key is shared or lost, the security of the data is no longer guaranteed. Critical actions for symmetric encryption methods represent the distribution, storage, and exchange of the keys. **Advanced Encryption Standard** (**AES**) and **Data Encryption Standard** (**DES**) are examples of symmetric encryption algorithms. This type of encryption is often used to encrypt large amounts of data, such as files on a hard drive or data sent over a network. **AES** is considered to be the most secure encryption algorithm nowadays.

Asymmetric Encryption

Asymmetric encryption, also known as **public-key encryption**, is a method of encryption that uses two different keys:

- a **public key**
- a **private key**

The public key is used to encrypt the data, while the private key is used to decrypt the data. This means anyone can use a public key to encrypt data for someone, but only the recipient with the associated private key can decrypt the data. Examples of asymmetric encryption methods include **Rivest–Shamir–Adleman** (**RSA**), **Pretty Good Privacy** (**PGP**), and **Elliptic Curve Cryptography** (**ECC**). Asymmetric encryption is used in a variety of applications, some of which include:

E-Signatures	SSL/TLS	VPNs
SSH	PKI	Cloud

Public-Key Encryption

One advantage of asymmetric encryption is its **security**. Since the security is based on very hard-to-solve mathematical problems, simple attacks cannot crack it. Furthermore, the issue of key exchange is bypassed. This is a significant problem with symmetric encryption methods. However, since the public key can be accessible to everyone, there is no need to exchange keys secretly. In addition, the asymmetric methods open up the possibility of authentication with digital signatures.

Data Encryption Standard

DES is a symmetric-key block cipher, and its encryption works as a combination of the one-time pad, permutation, and substitution ciphers applied to bit sequences. It uses the **same key** in both **encrypting and decrypting** data.

The key consists of **64 bits**, with **8 bits** used as a checksum. Therefore, the actual key length of DES is only **56 bits**. And that is why one always speaks of a key length of 56 bits when referring to DES. To prevent the danger from frequency analysis, not single letters, but each 64-bit block of plaintext is encrypted to a 64-bit block of ciphertext.

An extension of DES is the so-called **Triple DES / 3DES**, which encrypts data more securely. The procedure for this usually consists of three keys, with the first key being used to encrypt the data, the second to decrypt the data, and the third to encrypt the data again.

3DES was considered more secure than the original DES because it provides greater security using three rounds of encryption, although using a 56-bit key still limits it. **AES**, the successor to

DES, provides higher security using longer key lengths and is now the most widely used symmetric encryption technology.

Advanced Encryption Standard

Compared to DES, AES uses 128-bit (**AES-128**), 192-bit (**AES-192**), or 256-bit (**AES-256**) keys to encrypt and decrypt data. In addition, AES is faster than DES because it has a more efficient algorithm structure. This is because it can be applied to multiple data blocks at once, making it faster. This means that AES encryption and decryption can be performed faster than DES, which is especially important when large amounts of data need to be encrypted.

For example, we can find AES in many different applications and protocols, but they are not limited to:

WLAN IEEE 802.11i	IPsec	SSH
VoIP	PGP	OpenSSL

Cipher Modes

A cipher mode refers to how a block cipher algorithm encrypts a plaintext message. A block cipher algorithm encrypts data, each using fixed-size blocks of data (usually 64 or 128 bits). A cipher mode defines how these blocks are processed and combined to encrypt a message of any length. There are several common cipher modes, including:

Cipher Mode	Description
Electronic Code Book (ECB) mode	ECB mode is generally not recommended for use due to its susceptibility to certain types of attacks. Furthermore, it does not hide data patterns efficiently. As a result, statistical analysis can reveal elements of clear-text messages, for example, in web applications.
Cipher Block Chaining (CBC) mode	CBC mode is generally used to encrypt messages like disk encryption and e-mail communication. This is the default mode for AES and is also used in software like TrueCrypt, VeraCrypt, TLS, and SSL.
Cipher Feedback (CFB) mode	CFB mode is well suited for real-time encryption of a data stream, e.g., network communication encryption or encryption/decryption of files in transit like Public-Key Cryptography Standards (PKCS) and Microsoft's BitLocker.
Output Feedback (OFB) mode	OFB mode is also used to encrypt a data stream, e.g., to encrypt real-time communication. However, this mode is considered better for the data stream because of how the key stream is generated. We can find this mode in PKCS but also in the SSH protocol.
Counter (CTR) mode	CTR mode encrypts real-time data streams AES uses, e.g., network communication, disk encryption, and other real-time scenarios where data is processed. An example of this would be IPsec or Microsoft's BitLocker.
Galois/Counter (GCM) mode	GCM is used in cases where confidentiality and integrity need to be protected together, such as wireless communications, VPNs, and other secure communication protocols.

Each mode has its characteristics and is more suitable for certain use cases. The choice of encryption mode depends on the application's requirements and the security objectives to be achieved.



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